

Unit 4.1 Envisioning Sustainable Futures: Where do we want to go?

Up to this point we have studied development pathways, and what makes them more or less sustainable, from the analyst's seat. In Part 0 we looked at what makes sustainable development so hard. In Parts 1 and 2 we built an analytic framework for understanding how nature-society systems produce the pathways we observe, and in Part 3 we examined the capacities that societies need in order to reorient those pathways toward sustainability. Part 4 moves from the analyst's to the actor's seat. It asks what people pursuing sustainability can actually do. Turning concern into effective action requires informed agitators¹ who are willing and able to throw themselves into three tasks: i) to envision sustainable futures, ii) to identify the leverage points and policy instruments that can move development pathways toward those visions, and iii) to lead and mobilize others to act. We address these tasks in Units 4.1, 4.2, and 4.3; this unit takes up the first.

The teaching cases we have used in this course reconstruct development pathways looking backward: London across two millennia, Appalachia and Alaska across centuries. Advocates for sustainability face the opposite problem: they must imagine where they want development to go before a pathway there can be pursued. This is the practice Donella Meadows calls envisioning (reading 'a'): articulating, out loud and with others, what sustainable development should actually mean for a particular place and time. A vision in her sense is not a forecast of what is likely but a normative claim about what is worth wanting: for sustainable development, a claim about the fair distribution of improvements in well-being within and between generations (Unit 1.2). Meadows' argument is that the practice of envisioning sustainable futures is both essential and neglected. Neglected because too many advocates for sustainability start with diagnosing the unsustainability of current development pathways and then jump straight into rules, regulations, and technical fixes to address those immediate problems. This is understandable: immediate survival is the essential first step in the long term pursuit of sustainability. But not all immediate fixes lead toward pathways of sustainable development and, indeed, some may entail path-dependent lock-ins that make the pursuit of sustainability even more difficult. Envisioning sustainable futures is therefore essential to help assure that potential rules, regulations, and technical fixes are assessed not as ends in themselves but as means that may or may not contribute to the ends of both immediate relief *and* long term sustainability. It does this by creating shared visions of what sustainable development should look like in a particular place at a particular future time.

In Unit 3.7 we encountered the concept of sociotechnical imaginaries: collectively held visions, embedded in institutions, and performed in public. Shared visions of the future that reach this status are powerful because they produce not only a goal but the behavioral and institutional architecture for actors working within complex systems to actually move toward that goal. The goal of envisioning sustainable development is thus to develop visions vivid and persuasive enough to gain that kind of power: to organize budgets, legitimate policies, steer development

¹ "Informed agitation" is Amartya Sen's term for the combination of political mobilization and scientific understanding that the pursuit of sustainability requires; we introduced it in Unit 1.2 and returned to it in Unit 3.6. Sen, A. (2013). "The Ends and Means of Sustainability." *Journal of Human Development and Capabilities* 14(1): 6–20.

trajectories, and expand what people and institutions believe is possible. The film “A Message from the Future” (reading ‘b’) and the “seeds of a good Anthropocene” research program (reading ‘c’) are two contemporary attempts at this work, one through narrative and one through existing experiments.

Backcasting (reading ‘d’) connects a vision back to the present. Reason backward from the desired future: for the 2050 vision to remain reachable what must be true in 2040, in 2030, and next year? The answers turn a distant vision into near-term priorities.

To prepare for this unit, please review the following materials:

Preparation for class:

- a) **Read:** Meadows, Donella. (1994). “Envisioning a Sustainable World.” Address to the International Society for Ecological Economics, San José, Costa Rica.
<https://donellameadows.org/archives/envisioning-a-sustainable-world/>

Short and canonical; read it first.

- b) **Watch:** Klein, Naomi, & Molly Crabapple. (2019). “A Message from the Future with Alexandria Ocasio-Cortez.” *The Intercept*. <https://theintercept.com/2019/04/17/green-new-deal-short-film-alexandria-ocasio-cortez/> (Short film, about seven minutes.)

- c) **Read:** Bennett, Elena M., et al. (2016). “Bright spots: seeds of a good Anthropocene.” *Frontiers in Ecology and the Environment* 14(8): 441–448.
<https://doi.org/10.1002/fee.1309>

The journal charges for access at the link above. A free copy of the published article is available from the University of Southampton repository:

<https://eprints.soton.ac.uk/401154/1/fee1309.pdf>

- d) **Explore:** Government Office for Science. (2024). *The Futures Toolkit: Tools for Futures Thinking and Foresight Across UK Government*. London: HM Government.
<https://www.gov.uk/government/publications/futures-toolkit-for-policy-makers-and-analysts>

Work through the “Backcasting” tool only (direct link to that section: <https://www.gov.uk/government/publications/futures-toolkit-for-policy-makers-and-analysts/the-futures-toolkit-html#backcasting>); it is the how-to for study question IV. This is the methods manual used across the UK civil service, and backcasting is one of twelve tools in it: the others, from Horizon Scanning and Three Horizons to Scenarios, Visioning, and Roadmapping, are worth exploring.

Study Questions to help you get the most out of the readings:

- I. **Analyzing a Vision:** Watch “A Message from the Future” (reading ‘b’) as an exercise in the envisioning Meadows calls for in reading ‘a’. What vision of sustainable development does the film articulate, who is the “we” that shares it, and what would it take for the vision to move from circulating online to being embedded in institutions? Contrast it with the approach of Bennett et al. (reading ‘c’), which grows futures from existing “seeds”. What are the risks of each approach as a way of mobilizing action?
- II. **Returning to Goals:** In Unit 1.2 you explored the constituents of well-being, from your own life through the OECD's Better Life Index to the UN's global goals, and learned to distinguish ends from means. What does a vision of sustainable development do that a list of goals does not?
- III. **Your Case — Envision:** Articulate a vision for your case in Meadows' sense: out loud, on paper, and with others where you can. What would sustainable development actually mean for your place? Describe your case in 2050 on a pathway you would regard as sustainable. Be as concrete as possible, resist the urge to hedge, and say what you actually want for this place—whether it is your own home or the case you have chosen to devote your semester to working on.
- IV. **Your Case — Backcasting:** Using the backcasting method from reading 'd', connect the vision you articulated in question III back to the present: what would have to be true in 2040, in 2030, and in the next five years for your 2050 future to remain reachable? Which of those near-term changes could actors in your case start on now?

Digging deeper (optional materials for further exploring frontiers in the pursuit of sustainability):

- e) **Read:** Sharpe, Bill, Anthony Hodgson, Graham Leicester, Andrew Lyon, & Ioan Fazey. (2016). “Three Horizons: a pathways practice for transformation.” *Ecology and Society* 21(2): 47. <https://doi.org/10.5751/ES-08388-210247>

A practical method for connecting visions of the future to action in the present.

- f) **Read:** Johansson, Emma Li. (2021). “Participatory futures thinking in the African context of sustainability challenges and socio-environmental change.” *Ecology and Society* 26(4): 3. <https://doi.org/10.5751/ES-12617-260403>

How participatory envisioning is practiced on the ground, with a critical argument: consensus-seeking among mixed stakeholders can reproduce existing power relations, so marginalized groups may need to develop visions of their own first. Open access.